

Pandey & Kumar (2025) Mode-I Tensile Benchmark

Job: 1398386.mm.master02 (PK_MODE1_PROPOSED_PFM)

Provenance:

Job ID: 1398386.mm.master02

Mesh: NPHYS=14235, $h_{\min}=0.001$ mm

Material: $E=210$ GPa, $\nu=0.3$, $G_c=2.7$ kJ/m², $l_0=0.0075$ mm

Status: Terminated Step 2 Inc 3910 ($dt_{\min}=1.0E-9$ floor)

Reaction Force RF_y [kN]

1.0

0.8

0.6

0.4

0.2

0.0

Peak Load
 $RF_2 = 1.0810$ kN

Solver Cutback Termination
($dt < 1.0E-9$ s floor hit)
 $RF_2 = 0.9556$ kN

Proposed Adaptive Refined PFM (Abaqus UEL)

Peak Capacity: $RF_2 = 1.0810$ kN at $u = 8.8600$ μm

Last Converged (Inc 3909): $RF_2 = 0.9556$ kN at $u = 8.9041$ μm

0

2

4

6

8

Prescribed Displacement u_y [μm]